

Quiz (Carolina Reaper)

Q1. What is the solution to the inequality $|x| < 3$?

- (A) $(-3, 3)$
 (B) $[-3, 3]$
 (C) $(-\infty, -3) \cup (3, \infty)$
 (D) $(-\infty, 3] \cup [-3, \infty)$
 (E) $(-3, 0) \cup (0, 3)$

Q2. Solve the inequality $|2x - 5| \geq 3$

- (A) $x \leq -1$ or $x \geq 4$
 (B) $x \leq 1$ or $x \geq 4$
 (C) $x < -1$ or $x > 4$
 (D) $x < 1$ or $x > 4$
 (E) $x \geq -1$ and $x \leq 4$

Q3. Graph the solution set of $|x + 2| > 1$ on a number line

- (A) $(-\infty, -3) \cup (-1, \infty)$
 (B) $(-3, -1)$
 (C) $(-\infty, -1] \cup [3, \infty)$
 (D) $[-3, -1]$
 (E) $(-1, 3)$

Q4. The temperature in a city is never more than 25°C above or below 0°C . What inequality represents this situation?

- (A) $|T| \leq 25$
 (B) $|T - 25| \leq 0$
 (C) $|T| > 25$
 (D) $T \leq 25$
 (E) $T \geq -25$

Q5. The absolute value of x is less than 5. What is the solution in interval notation?

- (A) $(-5, 5)$
 (B) $[-5, 5]$
 (C) $(-\infty, -5) \cup (5, \infty)$
 (D) $(-\infty, 5] \cup [-5, \infty)$
 (E) $(-5, 0) \cup (0, 5)$

Q6. Solve $|3x| < 12$

- (A) $-4 < x < 4$
 (B) $-3 < x < 3$
 (C) $x > 4$ or $x < -4$
 (D) $x \geq -4$ and $x \leq 4$
 (E) $x < -4$ or $x > 4$

Q7. Solve the inequality $|x - 4| > 2$

- (A) $x < 2$ or $x > 6$
 (B) $x \leq 2$ and $x \geq 6$
 (C) $2 < x < 6$
 (D) $x \geq 2$ and $x \leq 6$
 (E) $x < 6$ or $x > 2$

Q8. What is the solution to $|2x + 1| \leq 5$?

- (A) $-3 \leq x \leq 2$
 (B) $-2 < x < 3$
 (C) $-2 \leq x \leq 3$
 (D) $x < -3$ or $x > 2$
 (E) $x \geq -2$ and $x \leq 3$

Open Ended Questions (FRQ)**Q9.** A company packages coffee in bags that weigh between 10 and 15 ounces. Using x as the weight of a bag of coffee in ounces, write an absolute value inequality that represents the range of weights.**Q10.** A child's height is restricted to be within 5 inches of 48 inches to be on a ride at an amusement park. Write an absolute value inequality to model the height restriction and solve it. Then explain what the solution represents in the context of the problem.

- Make sure students have calculators and pencils
- Provide extra paper if needed
- Keep an eye on the time and remind students to manage their time effectively